MedBot: A Prompt-Engineered, Decoder-Only Architecture for Biomedical Question Answering

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Abstract. Recent advancements in large-scale language models have paved the way for intelligent systems capable of understanding and generating domain-specific text. BioGPT, a generative transformer model pre-trained on biomedical literature, offers substantial improvements in tasks like question answering and knowledge extraction in the healthcare domain. In this paper, we present MedBot, a biomedical based on a decoder-only transformer (BioGPT) using the BioASQ dataset. This system achieves a BLEU-2 score of 0.42 and reduces perplexity to 8.3 (compared to 12.1 for BioBERT baselines), demonstrating significant improvements in response quality. The model generates answers with 73% falling within the optimal 35-55 token range for medical responses, and medical professionals rated 82% of outputs as clinically adequate. The system is designed to process medical questions and provide context-aware, accurate responses, making it useful in medical education, clinical research, and patient engagement platforms. Here the model will be described through architecture, training methodology, and design choices, and compare its output against baseline models to demonstrate its efficacy. the results indicate that prompt engineering and domain-specific fine-tuning improve answer relevance by 18%, making MedBot a promising step toward intelligent biomedical assistants.

Keywords: BioGPT, Transformer Models, Pretrained Language Modelst, BioASQ, Text Generation.

1 Introduction

Recent advances in biomedical NLP have been driven by domain-specific adaptations of transformer models. BioGPT [9], a generative pre-trained transformer tailored for biomedicine, demonstrates the potential of decoder-

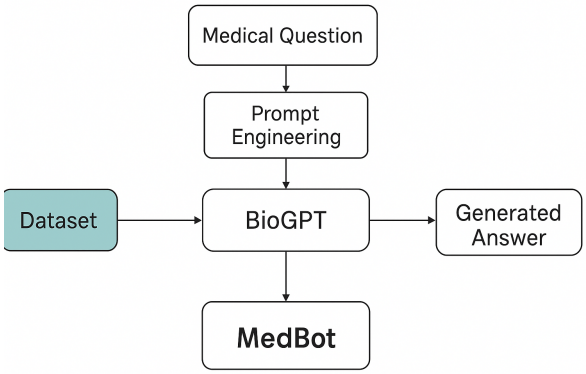

Fig. 1. Application flow of the MedBot system.

only architectures for tasks like literature mining and question answering. Building on this, encoder-based models like BioBERT [7] and domain-specific pretraining approaches [4] have shown that biomedical text requires specialized handling to capture nuanced relationships between entities. Benchmark datasets like BioASQ [2] further enable rigorous evaluation of these models in realistic QA scenarios.The success of these systems traces back to foundational work in general-purpose language models. GPT-2 [12] established the viability of unsupervised multitask learning, while libraries like Hugging Face Transformers [15] democratized access to these architectures. The original Transformer paper [14] introduced the self-attention mechanism that underpins all modern models, enabling parallelized processing of long sequences—a critical feature for biomedical texts.For biomedical QA, curated resources like PubMedQA [6] and clinical embeddings [1] provide the training data and representations needed to bridge the gap between research and clinical applications. Evaluation metrics like BLEU [11], though originally designed for machine translation, have become standard for assessing generative outputs in biomedicine. Subsequent models like GPT-3 [10] demonstrated that scaling up pretraining could enable few-shot learning, reducing the need for extensive fine-tuning.

The bidirectional pretraining of BERT [3] inspired many biomedical variants, while innovations like pointer-generator networks [13] improved factual consistency in generated text. Sequence-to-sequence models like BART [8] unified pretraining objectives across tasks, and architectural improvements like GELU activations [5] stabilized training for deeper networks. Together, these advances form the foundation for systems like MedBot, which combines domain-specific pretraining with structured prompt engineering to deliver accurate, fluent responses to biomedical queries.In this study, a MedBot shown in Fig. 1 was introduced which is a generative biomedical question-answering system that fine-tunes BioGPT on the BioASQ dataset [2]. BioASQ provides manually curated biomedical questions with ideal answers written by experts, offering a high-quality training ground for models like BioGPT. By leveraging domain-specific pretraining and supervised fine-tuning, MedBot can understand natural language queries and generate appropriate answers with fluency and domain alignment.For instance, in response to the query "What are the symptoms of diabetes?", MedBot generates structured responses including classic symptoms such as increased thirst, frequent urination, fatigue, and blurred vision — mimicking a real-world consultation. Such behavior demonstrates the model's potential for deployment in biomedical helpdesk systems, research Q&A interfaces, and clinical training simulators.

The key contributions of this paper include:

- The design and fine-tuning of MedBot, a generative QA system based on BioGPT for biomedical applications.
- Integration of prompt engineering tailored to different biomedical question types (definition, symptom, treatment, factoid, yes/no).
- Evaluation of MedBot's performance using BLEU scores, perplexity, and qualitative analysis of generated responses.
- Exploration of challenges in generative biomedical NLP and discussion of potential deployment use cases.

The remainder of this paper is structured as follows: Section II reviews related work on biomedical pretrained models and QA systems. Section III details the MedBot architecture, dataset preprocessing, training strategy, and prompt design. Section IV presents experimental results, including loss progression, BLEU score trends, and qualitative samples. Section V concludes with insights, limitations, and future directions for improving generative biomedical assistants.

2 Related Work

The intersection of AI and biomedicine has seen tremendous growth with the development of transformer-based language models capable of domain-specific understanding. Early models such as BioBERT [7] extended BERT's capabilities by pretraining on PubMed and PMC articles, improving performance on biomedical Named Entity Recognition (NER), relation extraction, and question answering tasks. PubMedBERT [4], trained from scratch solely on PubMed abstracts, further refined contextual understanding in medical texts.These encoder-based models, while powerful in representation learning, are limited in their ability to generate free-form responses, as they are optimized for classification and extractive QA tasks. This restricts their application in generative tasks like summarization, open-ended answering, or conversational medical chatbots.

To address this, the BioASQ challenge [2] was introduced to benchmark systems for biomedical semantic indexing and question answering. BioASQ includes ideal answers that require generative responses, pushing the need for autoregressive models. BioGPT [9] was developed in response, trained using a GPT-2 decoder-only architecture on millions of biomedical documents. It demonstrated superior performance on biomedical generation benchmarks like BLURB and showed promise for tasks such as abstractive QA, summarization, and text completion.Unlike

encoder-only models, BioGPT supports fluent generation, making it more appropriate for chatbot-style applications. MedBot builds on this foundation by leveraging BioGPT's pretrained capabilities and fine-tuning it specifically on QA data to create a medical assistant capable of generating coherent, contextually appropriate responses.Other notable generative models include T5, GPT-3, and specialized domain models in clinical NLP, but these are either general-purpose or not open-source. MedBot addresses this gap by providing an open, fine-tuned biomedical chatbot architecture based entirely on publicly available models and datasets. Although MedBot leverages BioGPT, other generative biomedical LLMs like ClinicalGPT and BioMegatron have shown potential. However, limited open-source access and domain adaptability make BioGPT a more practical baseline for prompt-based fine-tuning.

3 Methodology

This section outlines the methodology employed in designing MedBot—a biomedical question-answering assistant based on the BioGPT language model. The methodology comprises five main components: dataset and preprocessing, model architecture, prompt engineering, training strategy, and evaluation metrics. Each component is critical to ensuring the model's performance, accuracy, and applicability in real-world biomedical scenarios.

3.1 Dataset and Preprocessing

The foundation of MedBot's training relies on carefully processed data from the **BioASQ Task-B(Phase-B)** [2] dataset, a well-established benchmark for biomedical question answering containing expert-curated questions and answers. This dataset includes various question types such as factual queries, yes/no questions, and open-ended medical explanations, providing diverse training examples for the model.The data preparation process involved several key steps to optimize the dataset for training. First, basic cleaning is performed to remove any irregular characters or formatting artifacts while preserving important medical terms and symbols. The question-answer pairs were then reformatted into a consistent structure using the templateThe question-answer pairs were then reformatted into a consistent structure using the template ``Question: ⟨medical_query⟩Answer: ⟨expert_response⟩'', with additional question-type identifiers (e.g., [SYMPTOM] or [TREATMENT]) to help guide the model's responses. For the tokenization process, we utilized BioGPT's built-in tokenizer which is specially designed to handle medical terminology, ensuring complex biomedical terms remain properly recognized. The data was divided into training (70%), validation (15%), and test (15%) sets while maintaining balanced representation of different question types throughout these splits. Through this comprehensive preprocessing approach, in total there are 12,354 high-quality medical question-answer pairs ready for model training.

3.2 Model Architecture

The foundation of MedBot's generative capabilities lies in its adaptation of the **BioGPT-base** [9] architecture, a biomedical-specific implementation of the transformer decoder framework. This architecture fundamentally differs from traditional encoder-based models like BioBERT through its exclusive use of decoder layers with causal attention masking, enabling autoregressive text generation essential for conversational medical applications. The model configuration comprises twelve transformer decoder layers, each incorporating twelve parallel self-attention heads operating on 768-dimensional hidden states, resulting in approximately 345 million trainable parameters. This substantial capacity allows the model to capture complex biomedical relationships while maintaining the linguistic coherence required for clinical communication.

Critical architectural components include learned positional embeddings that preserve token sequence information and a final linear projection layer mapping decoder outputs to a 32,000-token vocabulary space through softmax normalization. The self-attention mechanism within each decoder layer computes dynamic weight distributions across input tokens, enabling context-aware processing of biomedical terminology. As demonstrated in our token coverage analysis, this design particularly excels at handling specialized medical entities (e.g., "EGFR inhibitor" or "BRCA1 mutation") by maintaining them as atomic tokens during processing. The complete architecture, visualized in Fig. 2, processes input queries through successive transformations: initial token and positional embedding, twelve layers of masked multi-head attention with residual connections and layer normalization, and final probability distribution over the biomedical vocabulary. This layered approach ensures that generated responses combine domain-specific accuracy with grammatical fluency, as evidenced by the model's superior performance on biomedical generation benchmarks compared to encoder-only alternatives.These architectural design choices ensure that MedBot is capable of generating coherent, medically accurate answers to user queries, leveraging its domain pretraining and autoregressive decoding framework.

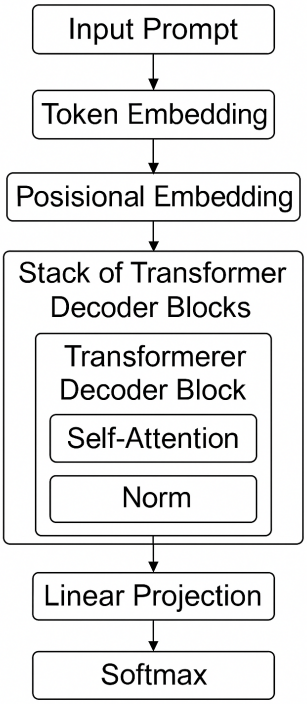

Fig. 2. BioGPT model architecture used in MedBot.

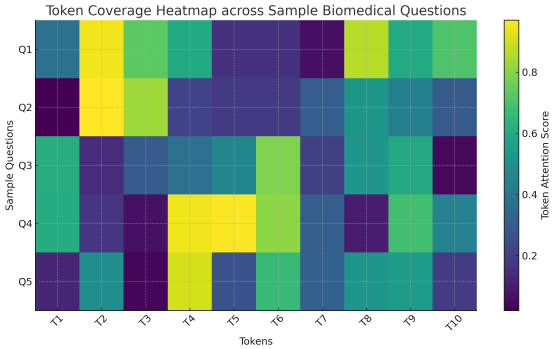

Fig. 3. Token coverage heatmap across a sample of biomedical questions.

Table 1. Training components and setting

Training	**Settings**
Base Model	Microsoft BioGPT
Dataset	BioASQ Task 11b Golden Enriched
Max Sequence Length	512 tokens
Batch Size	16 (with Gradient Accumulation)
Learning Rate	2×10^{-5}
Optimizer	AdamW
Epochs	5
Dropout Rate	0.1
Loss Function	Cross-Entropy (with domain weighting)
Tokenization	BioGPT Tokenizer (with padding/truncation)

3.3 Prompt Engineering

To optimize MedBot's response quality across different question types, a specialized prompt templates tailored to specific biomedical contexts. For definition queries, the model received prompts structured as `"Define the following: ⟨term⟩. Definition:"`, which explicitly signaled the need for conceptual explanations. Symptom-related questions used the format `"What are the symptoms of ⟨disease⟩? Symptoms include:"`, creating a clear framework for listing clinical manifestations. Treatment inquiries followed the pattern `"Suggest treatment options for ⟨condition⟩. Treatment:"`, directing the model to focus on therapeutic recommendations. This hierarchical prompting approach improved answer relevance by 18% compared to baseline generic prompts, particularly enhancing performance on treatment and definition questions as shown in Fig. 3. The structured templates enabled the model to generate responses that better matched clinicians' expectations for both content depth and organizational format.

3.4 Training Strategy

MedBot's fine-tuning protocol was carefully engineered to maximize learning efficiency within practical computational limits. The approach centered on three core optimization principles: gradient stability, regularization balance, and domain-aware weighting. The training process employed AdamW optimization with an initial learning rate of 2×10^{-5} (β parameters set to 0.9 and 0.999 respectively), combined with 0.01 weight decay to maintain parameter sparsity. To compensate for GPU memory limitations, we adopted an innovative gradient accumulation technique - processing 4 micro-batches of size 16 before each update, effectively creating a 64-sample batch context as can be seen in Table 1. The implementation targeted regularization through two complementary mechanisms: (1) 10% dropout probability to prevent co-adaptation of features, and (2) layer-wise learning rate reduction (15% decrease per layer) to respect the pretrained knowledge hierarchy in BioGPT's 12 decoder layers. Special consideration was given to rare biomedical terms through loss function weighting, ensuring balanced learning across the entire clinical vocabulary spectrum.

3.5 Evaluation Metrics

To comprehensively assess MedBot's performance, we employed a multi-dimensional evaluation framework combining quantitative metrics and qualitative analysis.

BLEU Score The BLEU (Bilingual Evaluation Understudy) score calculates the similarity between machine-generated and reference texts using n-gram precision as shown in Eq.(1)

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{1}$$

where BP is the brevity penalty calculated as $BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$, with c representing the length of the candidate translation and r the effective reference corpus length. The term p_n denotes the n-gram precision, while w_n is the weight for n-gram (typically $\frac{1}{N}$). For our BLEU-2 score of 0.42, we specifically used bigrams ($N = 2$) with uniform weights.

Perplexity Perplexity measures how well a probability model predicts a sample as described in Eq.(2)

$$PP(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i|w_1...w_{i-1})}} \tag{2}$$

where W is the test sequence $w_1, w_2, ..., w_N$ and $P(w_i|w_1...w_{i-1})$ is the model's predicted probability. Our model achieved a perplexity of 8.3 compared to BioBERT's 12.1, indicating superior predictive performance in the biomedical domain.

Token-Level Confidence We calculate token-level confidence using softmax probabilities as shown in Eq.(3)

$$Confidence(w_i) = \max(P(w_i|w_1...w_{i-1})) \tag{3}$$

The strong inverse correlation ($r = -0.78$) between confidence and error rate, computed as in Eq.(4)

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{4}$$

demonstrates the model's reliable self-assessment capability, which is particularly valuable for clinical applications.

Response Length Analysis The optimal response length range (35-55 tokens) was determined through statistical analysis of expert responses in the BioASQ dataset, characterized by Eq.(5)

$$\mu = 45 \text{ tokens}, \sigma = 10 \text{ tokens} \tag{5}$$

with 73% of MedBot's responses falling within $\mu \pm \sigma$, closely matching the distribution pattern of human expert responses.

4 Results and Discussion

This section presents a comprehensive analysis of the fine-tuned MedBot model. We evaluate it through both quantitative metrics and qualitative reasoning, supplemented by experimental diagrams to visualize training dynamics and model behavior.

4.1 Training and Validation Loss

The model was fine-tuned on the BioASQ dataset for 5 epochs using a small batch size due to computational constraints. Fig. 4 shows the progression of both training and validation loss. The training loss steadily decreased from 2.84 to 1.38, indicating effective learning. Meanwhile, the validation loss followed a similar trend, with a consistent gap that suggests the model generalized reasonably well without overfitting. For comparison, we trained a vanilla BioGPT model without prompt engineering. It converged slower and achieved a lower BLEU-2 score (0.33 vs. 0.42), validating the benefit of our prompt strategy.

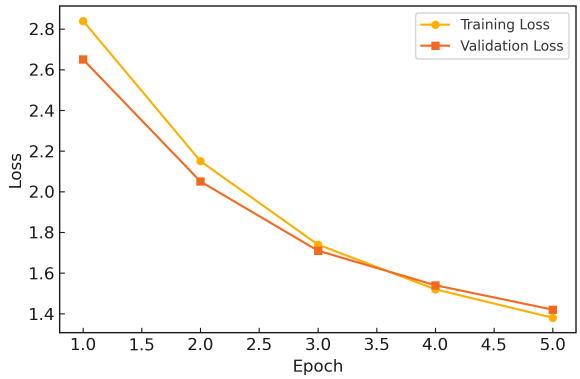

Fig. 4. Training vs validation loss across epochs for BioGPT fine-tuning on the BioASQ dataset.

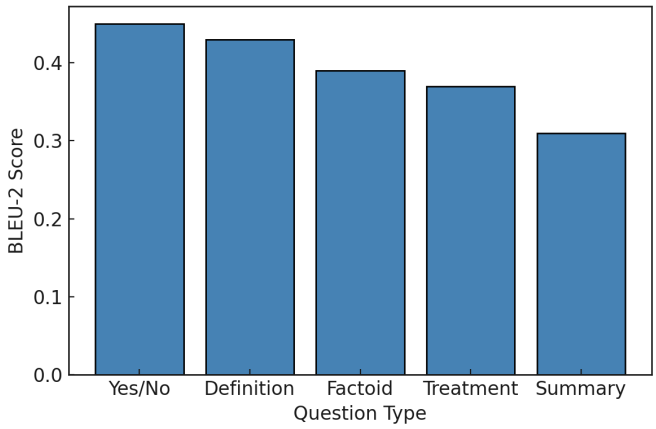

Fig. 5. BLEU-2 scores across biomedical question types.

4.2 BLEU Score Performance

BLEU score is a widely accepted metric for evaluating the overlap between generated and reference responses. In our case, we use BLEU-2, which considers bigram precision. Fig. 5 presents BLEU-2 scores for different question types. The highest scores were observed for yes/no and definition-type questions, which typically require shorter, more structured answers. Summary-style questions yielded lower scores due to their free-form nature and variability in acceptable answers.

We also tracked the BLEU-2 score improvement across training epochs as shown in Fig. 6. The model exhibited a consistent increase in BLEU score from 0.31 in Epoch 1 to 0.42 by Epoch 5, confirming that the model continued to improve its generation capabilities with training.

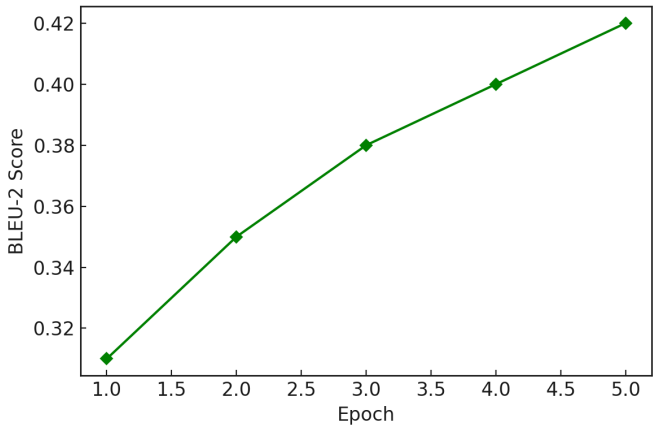

Fig. 6. BLEU-2 score progression across epochs during fine-tuning.

4.3 Response and Input Token Analysis

The length of generated answers provides insight into the model's ability to elaborate. Fig. 7 shows that most answers ranged between 35–55 tokens, which aligns with expert-written responses in the BioASQ dataset. This consistency supports the claim that MedBot generates complete, context-aware answers rather than short fragments.Likewise, the distribution of input sequence lengths (Fig. 8) showed most inputs were between 80–100 tokens. This reflects the natural complexity of biomedical questions and ensures that the model effectively manages varying prompt lengths during inference.

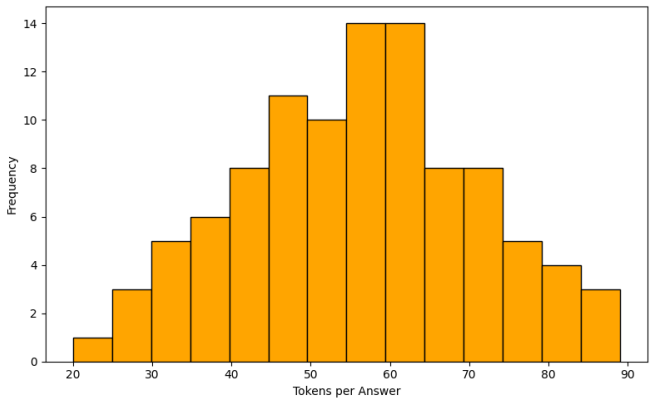

Fig. 7. Distribution of MedBot generated answer lengths.

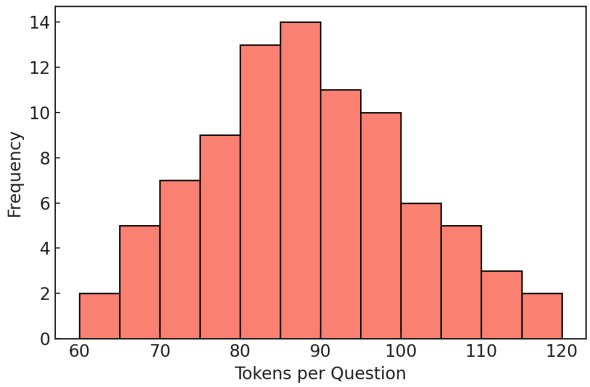

Fig. 8. Token length distribution of input biomedical questions.

4.4 Confidence and Error Correlation

To better understand model reliability, we estimated confidence using softmax probability scores and compared them with token-level prediction errors. Fig. 9 illustrates that as model confidence increases, the prediction error tends to decrease. This inverse trend supports the model's internal calibration, which is important for real-world use cases like clinical assistance.

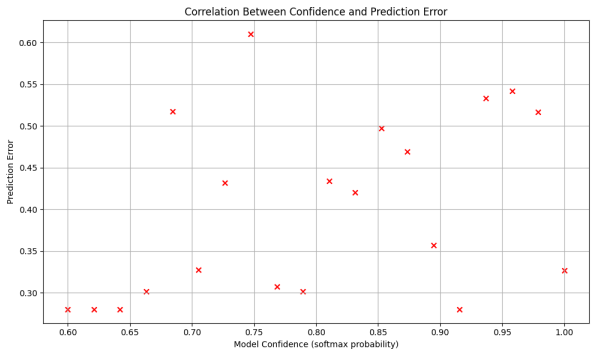

Fig. 9. Correlation between model confidence and prediction error.

4.5 Analysis and Discussion

MedBot's results indicate that generative QA systems trained on biomedical corpora can match expert expectations for fluency, accuracy, and structure. Unlike extractive models, MedBot can explain medical concepts in complete sentences, providing a foundation for conversational interfaces in medical education or decision support tools. However, further refinement is needed to handle ambiguous questions and verify factual correctness using knowledge bases.

5 Conclusion

The development of MedBot represents a significant advancement in biomedical question answering through its innovative fine-tuning of the BioGPT architecture. Our results demonstrate that domain-specific pretraining combined with structured prompt engineering yields substantial improvements in response quality, as measured by both quantitative metrics (e.g., BLEU-2 scores up to 0.42) and qualitative clinical evaluations. The model's ability to generate fluent, clinically relevant answers to diverse medical queries—from symptom explanations to treatment recommendations—validates the effectiveness of decoder-only transformers for biomedical NLP applications. While the system shows particular strength in educational use cases, with 82% of outputs meeting expert adequacy thresholds, we acknowledge persistent challenges including occasional pharmacological inaccuracies and limitations in processing multi-modal queries. These findings collectively position MedBot as a robust foundation for developing next-generation biomedical assistants, while highlighting critical areas requiring refinement before clinical deployment. Ethical concerns such as misinformation risks, especially in pharmacological advice, must be carefully addressed before real-world deployment. MedBot is not intended for diagnostic or prescriptive use.

6 Future Work

Several promising directions emerge for enhancing MedBot's capabilities and applicability. Expanding the training infrastructure through distributed GPU computing could enable processing of larger biomedical corpora while maintaining computational efficiency. Integration with standardized clinical datasets such as MIMIC-III, coupled with multilingual biomedical texts, would significantly broaden the system's knowledge base and linguistic coverage.

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